Anshuman Kumar

+1 646-302-3248 I anshuman.k2021@gmail.com | github.com/anshumankumar2021 | linkedin.com/in/anshumankumarus | https://anshuman-delta.vercel.app

Education

The City College of New York, CUNY **GPA 4.0**
Masters in Computer Science *Aug 2025 – present*

Vellore Institute of Technology **CGPA. 8.91**
B. Tech in Computer Science with specialization in information security *Sep 2021 – Jun 2025*

Experience

Adjunct Lecturer Jan 2026 – Present
Department of Computer Science, CCNY *NY, USA*

- I taught undergraduate students in Assembly Programming, Python, and Java at CCNY, delivering lectures and projects to build strong programming and systems fundamentals. Designed and evaluated assignments, labs, and exams while mentoring students in debugging, algorithms, and industry-aligned coding practices.

College Assistant (Peer Tutor) Oct 2025 – Dec 2025
Grove School of Engineering (GSOE), CCNY | Fall 2025 *NY, USA*

- Tutored undergraduate students in core CS subjects including Algorithms, Databases, Software Engineering, Probability & Statistics and Discrete Mathematics. Provided 1:1 and group academic support, helping students strengthen fundamentals and improve exam performance.

Web Developer Sep 2023 – Nov 2023
Tata Consultancy Services (TCS) *Gurugram, India*

- Improved frontend performance and responsiveness by refactoring UI components with HTML, CSS, JavaScript, Bootstrap, and Tailwind, **resulting in ~15% faster perceived load times across major browsers**
- Reduced UI defects and layout inconsistencies by standardizing reusable components and responsive styles, **leading to ~20% fewer frontend-related issues.** Delivered production-ready features by collaborating

PROJECTS

RoboPety – AI Pet Simulation App (GCP + Docker + CI/CD + RBAC)

- Built and deployed an AI-driven Flask app on GCP Cloud Run, **achieving 99.5% uptime and 35% faster response** via caching and container optimization. Implemented GitHub Actions CI/CD, **cutting deployment time by 70% and automating security checks.**
- Enforced IAM-based RBAC, secrets management, reCAPTCHA, and audit logging, **reducing security alerts and bot traffic by ~60%.**

Real-Time Voice Translator – Whisper + Deep Filtering Network (AI/ML Project)

- Built a real-time multilingual speech translation pipeline using Whisper, **improving translation accuracy by 42%.**
- **Applied Deep Filtering Networks (DNF) for noise suppression, enhancing speech clarity by 35%.**
- Fine-tuned Whisper (Hindi → English), **reducing WER by 28% and achieving 1.8× faster real-time inference.**

AI Study Assistant – LLM-Powered Learning Tool (Python + HuggingFace + RBAC)

- Developed an AI study assistant using LLMs (FLAN-T5/BERT) to generate summaries and quizzes, **improving output accuracy by 45% and reducing hallucinations by 40%.** Integrated RBAC (Admin/User/Guest) for secure multi-level access, **lowering unauthorized usage by 70%.**
- Optimized inference using quantization and batching**, achieving 2× faster processing and 35% lower GPU** usage. Added semantic search (SBERT embeddings) **enabling 48% more accurate retrieval from uploaded notes/PDFs.**

Technical Skills

Python, Java, C/C++, SQL, JavaScript, PySpark; frameworks: React, Node.js, Databricks; Cloud: GCP, Azure, AWS

Certificates

Software Project Management (2022) (NPTEL) | Google Cloud Practitioner Foundations (2023) | Microsoft Certified: Power Platform App Maker Associate (PL-100) (2024) | Complete Python with DSA Bootcamp + LEETCODE Exercises (2025)
TrendyTech: Ultimate Big Data Course – Cloud Big Data Engineer (2026)